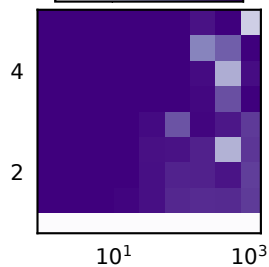


total harm

0.5 1.0

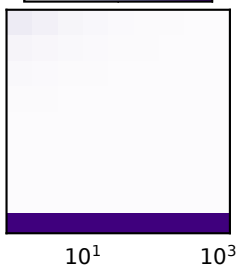
reproduction
number, R_0 

infection

0.5 1.0

4

2

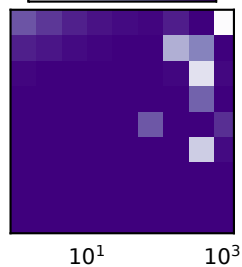


toxicity

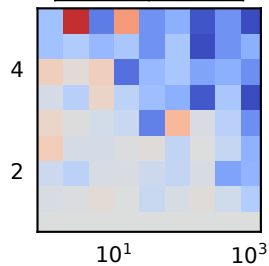
0.5 1.0

4

2

 ψ_{Ag}

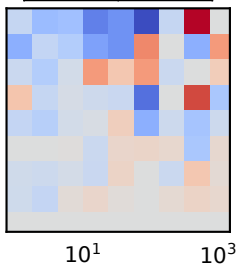
-2 0 2

reproduction
number, R_0  ψ_I

-2 0 2

4

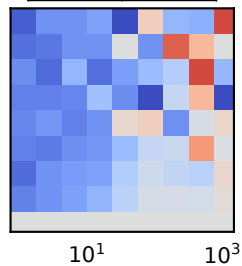
2

 ψ_E

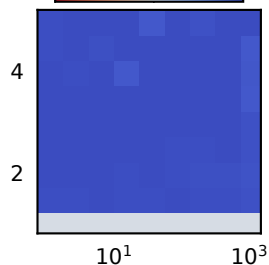
-2 0 2

4

2

 $g_{0,N \rightarrow N^*} = g_{0,E \rightarrow 2E}$

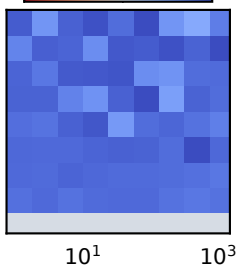
0.0 0.5

reproduction
number, R_0  $g_{0,N^* \rightarrow E}$

0.0 0.5

4

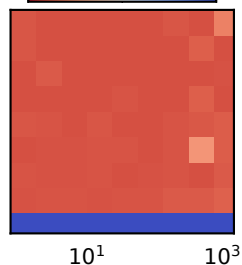
2

 $g_{0,E \rightarrow \emptyset}$

0.00 0.25 0.50

4

2

immunogenicity, κ immunogenicity, κ immunogenicity, κ